# Vulnerability Assessment Report

Cybersecurity Group C

Prepared by Group C Tester · 6 November 2026

Overall risk: HIGH (45.7/100)

# 1. Executive Summary

The assessment identified 61 vulnerabilities across 8 scanned hosts in the environment, resulting in an overall risk rating of HIGH (risk index 45.7/100). Three Critical findings require immediate remediation due to their potential impact on business operations. 7 High severity vulnerabilities represent elevated risk and should be addressed in accordance with organizational risk management policies. A further 12 Medium and 15 Low severity issues were observed, along with 24 informational items that support asset and configuration visibility. It is recommended that remediation efforts prioritise externally exposed systems and unsupported software, followed by a verification scan to confirm closure.

# 2. Scope

The assessment covered internal production servers, network infrastructure and externally exposed services provided by the client. Authenticated and unauthenticated scans were performed using Tenable Nessus.

# 3. Methodology

Findings were collected via automated vulnerability scanning, normalised, de-duplicated and rated using CVSS base scores. Risk was calculated using weighted severity points (Critical 10, High 7, Medium 5, Low 2) and mapped to an overall risk level.

# 4. Findings Summary

| Severity | Count | Points |
|---|---|---|
| Critical | 3 | 30 |
| High | 7 | 49 |
| Medium | 12 | 60 |
| Low | 15 | 30 |
| Info | 24 | 0 |

# 5. Detailed Findings

| ID | Vulnerability | Severity | CVSS | Affected System | Status |
|---|---|---|---|---|---|
| VR-001 | Ubuntu 16.04 End of Life | Critical | 10.0 | web-prod-01.corp.local (10.10.4.21) | Open |
| VR-002 | Apache Log4j Remote Code Execution (Log4Shell) | Critical | 10.0 | dc-01.corp.local (10.10.1.10) | Open |
| VR-003 | Microsoft Windows SMBv1 Multiple Vulnerabilities (MS17-010) | Critical | 9.8 | jump-host-01.corp.local (10.10.9.2) | In Progress |
| VR-007 | Remote Desktop Protocol BlueKeep | High | 8.8 | file-srv-01.corp.local (10.10.2.40) | Open |
| VR-005 | PHP 7.4 Unsupported Version | High | 8.6 | mail-relay-01.corp.local (10.10.3.8) | Remediated |
| VR-004 | OpenSSH Multiple Vulnerabilities | High | 8.1 | app-api-03.corp.local (10.10.4.33) | In Progress |
| VR-010 | Windows Print Spooler Privilege Escalation | High | 7.8 | jump-host-01.corp.local (10.10.9.2) | Open |
| VR-006 | MySQL 5.7 Multiple Vulnerabilities | High | 7.5 | vpn-gw-01.corp.local (172.16.0.5) | Open |
| VR-008 | Apache HTTP Server 2.4 Request Smuggling | High | 7.5 | web-prod-01.corp.local (10.10.4.21) | In Progress |
| VR-009 | SNMP Agent Default Community Name (public) | High | 7.5 | dc-01.corp.local (10.10.1.10) | Remediated |
| VR-012 | SSL Certificate Cannot Be Trusted | Medium | 6.5 | db-core-02.corp.local (10.10.6.14) | Remediated |
| VR-013 | TLS Version 1.0 Protocol Detection | Medium | 6.5 | app-api-03.corp.local (10.10.4.33) | In Progress |
| VR-018 | SSL Certificate Cannot Be Trusted | Medium | 6.5 | dc-01.corp.local (10.10.1.10) | In Progress |
| VR-019 | TLS Version 1.0 Protocol Detection | Medium | 6.5 | jump-host-01.corp.local (10.10.9.2) | Open |
| VR-014 | SSL Medium Strength Cipher Suites (SWEET32) | Medium | 5.9 | mail-relay-01.corp.local (10.10.3.8) | In Progress |
| VR-020 | SSL Medium Strength Cipher Suites (SWEET32) | Medium | 5.9 | db-core-02.corp.local (10.10.6.14) | Remediated |
| VR-011 | HTTP Information Disclosure | Medium | 5.3 | jump-host-01.corp.local (10.10.9.2) | Accepted |
| VR-015 | SMB Signing Not Required | Medium | 5.3 | vpn-gw-01.corp.local (172.16.0.5) | Accepted |
| VR-016 | Nginx Version Disclosure | Medium | 5.3 | file-srv-01.corp.local (10.10.2.40) | Remediated |
| VR-017 | HTTP Information Disclosure | Medium | 5.3 | web-prod-01.corp.local (10.10.4.21) | Open |
| VR-021 | SMB Signing Not Required | Medium | 5.3 | app-api-03.corp.local (10.10.4.33) | Open |
| VR-022 | Nginx Version Disclosure | Medium | 5.3 | mail-relay-01.corp.local (10.10.3.8) | Remediated |
| VR-027 | X.509 Certificate Subject CN Mismatch | Low | 3.1 | mail-relay-01.corp.local (10.10.3.8) | Accepted |
| VR-032 | X.509 Certificate Subject CN Mismatch | Low | 3.1 | jump-host-01.corp.local (10.10.9.2) | In Progress |
| VR-037 | X.509 Certificate Subject CN Mismatch | Low | 3.1 | file-srv-01.corp.local (10.10.2.40) | Accepted |
| VR-023 | SSH Server CBC Mode Ciphers Enabled | Low | 2.6 | dc-01.corp.local (10.10.1.10) | In Progress |
| VR-025 | Web Server Allows Password Auto-Completion | Low | 2.6 | db-core-02.corp.local (10.10.6.14) | Open |
| VR-026 | SSH Weak MAC Algorithms Enabled | Low | 2.6 | app-api-03.corp.local (10.10.4.33) | Accepted |
| VR-028 | SSH Server CBC Mode Ciphers Enabled | Low | 2.6 | vpn-gw-01.corp.local (172.16.0.5) | Open |
| VR-030 | Web Server Allows Password Auto-Completion | Low | 2.6 | web-prod-01.corp.local (10.10.4.21) | Accepted |
| VR-031 | SSH Weak MAC Algorithms Enabled | Low | 2.6 | dc-01.corp.local (10.10.1.10) | Open |

| ID | Vulnerability | Severity | CVSS | Affected System | Status |
|---|---|---|---|---|---|
| VR-033 | SSH Server CBC Mode Ciphers Enabled | Low | 2.6 | db-core-02.corp.local (10.10.6.14) | Remediated |
| VR-035 | Web Server Allows Password Auto-Completion | Low | 2.6 | mail-relay-01.corp.local (10.10.3.8) | In Progress |
| VR-036 | SSH Weak MAC Algorithms Enabled | Low | 2.6 | vpn-gw-01.corp.local (172.16.0.5) | Open |
| VR-024 | ICMP Timestamp Request Remote Date Disclosure | Low | 2.1 | jump-host-01.corp.local (10.10.9.2) | Open |
| VR-029 | ICMP Timestamp Request Remote Date Disclosure | Low | 2.1 | file-srv-01.corp.local (10.10.2.40) | In Progress |
| VR-034 | ICMP Timestamp Request Remote Date Disclosure | Low | 2.1 | app-api-03.corp.local (10.10.4.33) | Remediated |

# 6. Risk Analysis

Weighted risk points: 169. Risk index: 45.7/100. Overall risk level: HIGH. Critical: 3, High: 7, Medium: 12, Low: 15, Informational: 24.

# 7. Recommendations

| Finding | Severity | Recommendation |
|---|---|---|
| Ubuntu 16.04 End of Life | Critical | Upgrade to a supported Ubuntu Long-Term Support (LTS) version and apply all security updates. |
| Apache Log4j Remote Code Execution (Log4Shell) | Critical | Patch the web server, suppress version banners and remove verbose error and directory listing output. |
| Microsoft Windows SMBv1 Multiple Vulnerabilities (MS17-010) | Critical | Disable SMBv1, require SMB signing and restrict SMB exposure to trusted network segments. |
| Remote Desktop Protocol BlueKeep | High | Enable Network Level Authentication, restrict RDP behind VPN and apply the latest Windows updates. |
| PHP 7.4 Unsupported Version | High | Upgrade to a supported Ubuntu Long-Term Support (LTS) version and apply all security updates. |
| OpenSSH Multiple Vulnerabilities | High | Update OpenSSH to the latest supported version and disable unnecessary access methods such as password and root login. |
| Windows Print Spooler Privilege Escalation | High | Apply the latest Microsoft cumulative security update through your patch management process. |
| MySQL 5.7 Multiple Vulnerabilities | High | Patch the database engine, restrict network access to application hosts and enforce strong authentication. |
| Apache HTTP Server 2.4 Request Smuggling | High | Patch the web server, suppress version banners and remove verbose error and directory listing output. |
| SNMP Agent Default Community Name (public) | High | Change default community strings, migrate to SNMPv3 and restrict SNMP to management hosts. |
| SSL Certificate Cannot Be Trusted | Medium | Disable deprecated protocols (SSLv3, TLS 1.0/1.1), remove weak ciphers and deploy certificates from a trusted CA. |
| TLS Version 1.0 Protocol Detection | Medium | Disable deprecated protocols (SSLv3, TLS 1.0/1.1), remove weak ciphers and deploy certificates from a trusted CA. |
| SSL Medium Strength Cipher Suites (SWEET32) | Medium | Disable deprecated protocols (SSLv3, TLS 1.0/1.1), remove weak ciphers and deploy certificates from a trusted CA. |
| HTTP Information Disclosure | Medium | Patch the web server, suppress version banners and remove verbose error and directory listing output. |
| SMB Signing Not Required | Medium | Disable SMBv1, require SMB signing and restrict SMB exposure to trusted network segments. |
| Nginx Version Disclosure | Medium | Patch the web server, suppress version banners and remove verbose error and directory listing output. |
| X.509 Certificate Subject CN Mismatch | Low | Disable deprecated protocols (SSLv3, TLS 1.0/1.1), remove weak ciphers and deploy certificates from a trusted CA. |
| SSH Server CBC Mode Ciphers Enabled | Low | Update OpenSSH to the latest supported version and disable unnecessary access methods such as password and root login. |
| Web Server Allows Password Auto-Completion | Low | Patch the web server, suppress version banners and remove verbose error and directory listing output. |
| SSH Weak MAC Algorithms Enabled | Low | Update OpenSSH to the latest supported version and disable unnecessary access methods such as password and root login. |
| ICMP Timestamp Request Remote Date Disclosure | Low | Review the finding, apply vendor patches or configuration hardening, and validate with a rescan. |

## 8. Conclusion

The environment currently presents a HIGH level of risk. Addressing Critical and High findings within the recommended timeframes and establishing a recurring scan cadence will materially reduce the organisation's attack surface.